

Project – Employee Performance

1. Project Overview :-

The dataset contains **500 employee records** across five departments: Marketing, Finance, IT, HR, and Sales.

It tracks employee demographics, compensation, performance, training, and attrition. This is a **workforce analytics dataset** suitable for HR insights, attrition prediction, and performance evaluation.

2. Data Structure :-

Columns :-

- EmployeeID - Unique identifier for each employee .
- Department - Department (Marketing, Finance, IT, HR, Sales).
- JobRole - Specific role (Analyst, Developer, Manager, Executive, etc).
- Joining Date - Excel serial date format (needs conversion to actual date).
- Salary - Annual salary (numeric).
- Annual salary (numeric) - Performance rating (1–9 scale) .
- Training Hours - Number of training hours completed.
- AttritionFlag - Yes/No indicator of whether the employee left .

3. Derived column to create :-

- **Tenure (Years)** = Current Date – JoiningDate (convert Excel serial to actual date).
- **Salary Band** = Categorize salaries into ranges (Low, Medium, High).
- **Performance Category** = Group scores (e.g., 1–3 = Low, 4–6 = Medium, 7–9 = High).
- **Training Effectiveness** = PerformanceScore ÷ TrainingHours (to measure ROI of training).
- **Attrition Risk Index** = Weighted score combining Salary, Performance, Training, and AttritionFlag.
- **Department Attrition Rate** = Attrition count ÷ Department size.
- **Average Salary per Role** = Useful for compensation benchmarking.

4. Key Insights :-

- **Attrition is spread across all departments**, but Sales and Marketing show higher attrition flags compared to IT and Finance.
- **Low performance scores (1–3)** often correlate with attrition (e.g., EmployeeID 3, 12, 20, 21).
- **High training hours don't always prevent attrition** — some employees with >40 hours still left (e.g., EmployeeID 15, 22).

- **Salary is not a strong predictor alone** — both high and low salary employees have attrition flags.
- **Managers and Executives show mixed attrition trends** — some leave despite high salaries and performance, suggesting role stress or mismatch.
- **Finance department** has several high performers (scores 8–9) with no attrition, indicating stability compared to Marketing/Sales.

❖ Client Questions :-

1. Which department has the highest average performance score ?

Solution :-

- a) First import csv file in python jupyter notebook .
- b) Second use import pandas library in python.
- c) Third extract department and performance score from main data file .
- d) Then use grouping function and tranfer that result in system memory and import in power bi.

	A	B	C	D	E	F	G	H
1	Employee_ID	Department	Job_Role	Joining_Date	Salary	Performance_Score	Training_Hours	Attrition_Flag
2	1	Marketing	Analyst	17-05-2018 0.00	70603	8	34	Yes
3	2	Marketing	Developer	01-06-2017 0.00	106708	4	3	No
4	3	Marketing	Executive	19-08-2017 0.00	111926	1	4	No
5	4	Finance	Executive	11-07-2020 0.00	98803	1	0	No
6	5	IT	Manager	14-11-2022 0.00	69332	4	3	No
7	6	Marketing	Developer	12-04-2018 0.00	51312	6	11	No
8	7	Marketing	Analyst	06-03-2017 0.00	114526	8	26	No
9	8	Finance	Developer	15-12-2016 0.00	74179	6	41	No
10	9	Finance	Executive	18-09-2015 0.00	101442	9	22	No
11	10	Marketing	Executive	10-12-2024 0.00	70379	9	24	No
12	11	HR	Manager	24-06-2015 0.00	47297	2	20	Yes
13	12	Marketing	Manager	14-10-2019 0.00	32011	1	23	No
14	13	Marketing	Executive	31-01-2020 0.00	74014	2	24	Yes
15	14	Finance	Analyst	23-02-2019 0.00	41738	5	44	No
16	15	Finance	Analyst	22-12-2024 0.00	77498	4	45	Yes
17	16	Marketing	Developer	19-01-2024 0.00	41120	4	31	No
18	17	Marketing	Executive	02-05-2024 0.00	38767	9	31	No
19	18	Marketing	Executive	21-12-2024 0.00	52852	8	4	No
20	19	Finance	Analyst	17-01-2024 0.00	109239	6	35	Yes
21	20	Marketing	Analyst	21-04-2018 0.00	30929	1	45	Yes
22	21	Marketing	Executive	18-10-2022 0.00	107487	2	41	Yes
23	22	Sales	Manager	22-12-2024 0.00	47438	6	48	Yes
24	23	HR	Executive	18-07-2018 0.00	78515	2	41	No
25	24	IT	Executive	26-04-2016 0.00	35744	5	38	No
26	25	HR	Analyst	12-05-2023 0.00	113820	5	15	No
27	26	Sales	Analyst	19-08-2017 0.00	72417	6	32	Yes
28	27	IT	Manager	03-09-2019 0.00	37069	6	48	No
29	28	Marketing	Manager	23-01-2017 0.00	72359	6	20	No
30	29	Sales	Developer	03-08-2020 0.00	49799	2	8	No
31	30	Marketing	Executive	17-07-2020 0.00	100188	3	5	No
32	31	Finance	Analyst	05-10-2020 0.00	119859	6	44	No
33	32	Sales	Manager	27-04-2019 0.00	37597	8	34	Yes

```
[147] import pandas as pd

[148] # For the path for manually typing it without copy-pasting
# or by removing the backslash character
x = pd.read_csv("C:\Users\paulo\Desktop\deatlog\src\150employee_performance.csv")
# Alternative approach using forward slashes which works across operating systems
# x = pd.read_csv("C:\Users\paulo\Desktop\deatlog\src\150employee_performance.csv")

[149]
Employee_ID Department Job_Role Joining_Date Salary Performance_Score Training_Hours Attrition_Flag
0 1 Marketing Analyst 2018-05-17 00:00:00 70603 8 34 Yes
1 2 Marketing Developer 2017-06-01 00:00:00 106708 4 3 No
2 3 Marketing Executive 2017-08-19 00:00:00 111926 1 4 No
3 4 Finance Executive 2020-07-11 00:00:00 98803 1 0 No
4 5 IT Manager 2022-11-14 00:00:00 69332 4 3 No
... ..
495 496 Sales Manager 2019-09-28 00:00:00 42785 2 30 No
496 497 Marketing Executive 2021-08-08 00:00:00 75471 7 7 No
497 498 HR Developer 2019-05-12 00:00:00 119888 3 14 No
498 499 Sales Analyst 2022-06-18 00:00:00 118880 4 30 No
499 500 Finance Executive 2021-07-11 00:00:00 85751 3 29 No

500 rows x 8 columns

[150] x[x['Department'] == 'Performance_Score']
x
[151]
Department Performance_Score
0 Marketing 8
1 Marketing 4
2 Marketing 1
3 Finance 1
4 IT 4
... ..
495 Sales 2
496 Marketing 7
497 HR 3
498 Sales 4
499 Finance 1

500 rows x 2 columns

[152] x[x.groupby('Department')['Performance_Score'].mean().round(0)]
x
[153]
Performance_Score
Department
Finance 4.0
HR 5.0
IT 5.0
Marketing 5.0
Sales 5.0

[154] x[x.read_csv("C:\Users\paulo\Desktop\deatlog\src\150.csv")]
pd.Series("success")
```

2. Which 6 employees performance outstanding in sales ?

Solution :-

- a) First import csv file in python jupyter notebook .
- b) Second use import pandas library in python.
- c) Third extract employee id and salary from main data file .

- d) Then use grouping function and transfer that result in system memory and import in power bi.

```
[16]: c4=x[["Employee_ID","Salary"]]
      x3=c4.groupby('Employee_ID').sum().reset_index().sort_values("Salary",ascending=False)
      x3
```

```
[16]:
```

	Employee_ID	Salary
342	343	119963
30	31	119859
246	247	119833
301	302	119678
274	275	119571
...	...	...
84	85	30358
78	79	30340
189	190	30276
277	278	30159
83	84	30138

500 rows x 2 columns

```
[17]: x3.to_csv(r'C:\Users\yashm\OneDrive\Desktop\p3\ 2nd.csv')
      print('success')
      success
```

3. What is the attrition rate by department ?

Solution :-

- First import csv file in python jupyter notebook .
- Second use import pandas library in python.
- Third extract attrition flag that equal to Yes second step count how much attrition department in that column **{third divide attrituioun yes result with attrition department count values}** from main data file .
- Then tranfer that final result in system memory and import in power bi.

```
[18]: attrition = x[x['Attrition_Flag'] == 'Yes']
      attrition_counts = attrition['Department'].value_counts()
      dept_counts = x['Department'].value_counts()
      attrition_rate = (attrition_counts / dept_counts * 100).round(1)
      attrition_rate
```

```
[18]: Department
      Finance    18.6
      HR         26.3
      IT         16.1
      Marketing  16.7
      Sales      23.1
      Name: count, dtype: float64
```

```
[19]: attrition_rate.to_csv(r'C:\Users\yashm\OneDrive\Desktop\p3\ 3rd.csv')
      print('success')
      success
```

4. Which 5 department perform well in performance score ?

Solution :-

- First import csv file in python jupyter notebook .
- Second use import pandas library in python.
- Third extract Department and performance score from main data file .
- Then use grouping function and transfer that result in system memory and import in power bi.

```
[58]: z1=x[['Department','Performance_Score']]
      z1.groupby('Department').sum().sort_values('Performance_Score',ascending=False).reset_index()
```

```
[58]:
```

	Department	Performance_Score
0	Sales	552
1	Marketing	551
2	HR	461
3	IT	455
4	Finance	428

```
[59]: z1.to_csv(r'C:\Users\yashm\OneDrive\Desktop\p3\ 4th.csv')
      print('success')
```

POWER BI REPORT :-



